

Gautham Kailash

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EXPERIENCE

Squarepoint Capital

Jul. 2026 – Present

C++ Software Engineer (Risk PnL)

Singapore

- Designed a state machine processing 50-60M trades/day to maintain live order and trade state from streaming data.
- Built a codegen tool to derive SQL schemas from protobuf, and led migration of Risk databases to support JSON columns.

Squarepoint Capital

Aug. 2025 – Jun. 2026

C++ Software Engineer Intern (Risk PnL)

Singapore

- Built ingestors feeding real-time data into a high-availability distributed system for multi-asset trade booking and PnL.
- Added support for nested types in the ingestion layer to handle more complex instrument data.
- Implemented real-time C++ auto-marking of instruments, processing positions 400× faster for reliable PnL calculations.
- Decoupled a gRPC service into a standalone binary, eliminating resource contention and speeding up data lookups.
- Replaced special-case end-of-day roll logic with a unified roll client, automating a previously manual, error-prone process.

Goldman Sachs

May 2025 – Aug. 2025

Software Engineer Intern (Margin Engineering)

Singapore

- Designed and implemented a job orchestration framework, enabling dependency management and extensible scheduling.
- Engineered a real-time platform to process daily margin events, handling millions of rows and replacing manual queries.

AI Centre for Education Technologies

May 2024 – Aug. 2024

Software Engineer Intern

Singapore

- Built core features for Koditsu, a code-assessment platform engineered to handle 800+ concurrent users under load.

EDUCATION

National University of Singapore

Aug. 2022 – May 2026

BSc. Computer Science, GPA: 4.52 / 5.00 (First Class Honours, Highest Distinction)

Singapore

- **Awards:** [VIP Grant](#) (\$10,000) | [Dean's List](#) | [NUS College President's Honour Roll](#) | SINDA Excellence Award
- **Teaching:** CS2030S Programming Methodology II, CS2105 Computer Networks
- **Leadership:** VP of Pre-100 Club, Faculty Coordinator for NUSC Mentorship Programme, Co-Founder of [Voila](#)
- **Courses:** Computer Networks, Operating Systems, Parallel and Concurrent Programming, Distributed Algorithms

PROJECTS

Torus | <https://github.com/kailashgautham/torus> | C++

- Developed a lock-free SPSC queue in C++23 doing ~950M ops/sec, 79% of boost::lockfree's throughput.
- Removed false sharing with cache-line padding, replaced modulo with bitmask indexing, validated with ThreadSanitizer.

Lox Interpreter | <https://github.com/kailashgautham/lox-interpreter> | C++

- Developed a functional Lox language interpreter in C++.

KaithanOS | <https://github.com/kaithanos/kaithanos> | C, Assembly

- Developed and implemented an OS bootloader in C, ensuring efficient initial hardware and system software loading.
- Integrated the FAT file system into the OS, enabling disk read/write capabilities and supporting FAT subdirectories.
- Engineered interrupt handling with IDT/GDT and optimised memory management for enhanced performance.

Concurrent Systems | <https://github.com/kailashgautham/concurrent-systems> | C++, Go, Rust

- **C++, Go:** Developed a high-performance exchange matching engine, handling up to 100 threads and 4 million orders.
- **Rust:** Created a multi-client TCP server with Tokio for async task management, achieving a 30x speedup.

TECHNICAL SKILLS

- **Languages:** C++, C#, C, Java, Python
- **Frameworks:** CUDA, OpenMP, Boost, gRPC, Protobuf
- **Developer Tools:** Git, Vim, Unix, Bash